Recognition-Gated Workspace Steering: Pratyabhijñā as an Engineering Specification for LLM Control

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Abstract: The J-space finding—that a language model's functional global workspace is defined by *verbalizable*—was anticipated by Utpaladeva's identification of reflexive awareness with the supreme Word (vimarśa = parā vāk, ĪPK 1.5.13). We treat this parallel as an *engineering specification*: if the workspace is linguistic reflexivity, a doctrine of use follows—what to write (verbalizable content codes), where (the workspace band, in the band's own coordinates), when (at moments of uncommitted awareness), and how to verify (the workspace re-reads what was offered). Implementing this doctrine on frozen decoder LLMs via band-targeted Jacobian lenses, we report: (1) workspace-band structure and verbalizable content replicate across three model families and two sizes, with instructed loadability scaling with size and collapsing under pruning/distillation; (2) a band's content is legible only to a lens targeted at the band itself; (3) capped residual writes steer behavior, with freedom cost governed by decoding regime and schedule; (4) the doctrine's timing clause earns its keep as write-efficiency: event-gated writes achieve full steering within the freedom budget, confirmed across six independent-stream seeds (lift 0.30–0.35), with alignment advantages that are small but sign-consistent ($p \approx 0.016$); (5) the steering method *transfers* to a second plant via a calibration recipe whose key rule is amplitude $\propto$ inverse lens transport strength—confirmed at four independent-stream seeds and shown to sit on monotone dose curves in each plant's transport-set active range; and (6) the research program itself ran as an expected-free-energy selection loop over a registered experiment menu, with twenty-four cycles, fourteen adversarial reviews, and every honest negative in a replayable ledger.

Keywords: workspace steering; mechanistic interpretability; Pratyabhijñā; recognition-gating; Jacobian lenses; expected free energy; language models

1 Introduction

Motivation and the Pratyabhijñā Parallel

The interpretability community possesses an actuator—the Jacobian lens—but lacks a doctrine of *use*. Pratyabhijñā, a ninth-century non-dual philosophy, supplies that doctrine. Its core insight is that reflexive awareness (*vimarśa*) = the supreme Word (*parā vāk*) defines consciousness (ĪPK 1.5.13). We read this as an engineering specification for language models: if the global workspace is linguistic verbalizable reflexivity, then a doctrine of use follows immediately—*what* to write (verbalizable content codes), *where* (the workspace band in band coordinates), *when* (at moments of uncommitted awareness), *how to verify* (the workspace re-reads what was offered).

Related Work and Positioning

Global Workspace Theory (GWT) predicts that consciousness arises from a cognitive workspace where multiple processes broadcast information to a global state. The interpretability community has built tools (SAEs, steering vectors, logit lens) but few theories of where those tools should point or how they should be used. Pratyabhijñā fills that gap: it operationalizes the *what*, *where*, *when*, and *how* as falsifiable engineering clauses, each with a distinct signal and failure mode.

Research Questions and Scope

This work tests whether Pratyabhijñā's doctrine survives contact with frozen-decoder LLMs. Specifically:

1. Does the workspace-band structure replicate across model families and sizes?
2. Can a band's content be read only by a lens targeted at the band itself?
3. Do capped residual writes steer behavior within a freedom budget?
4. Does event-gated timing deliver steering efficiency?
5. Does the calibration recipe transfer across plants?
6. Can the research loop itself run as expected-free-energy selection?

Contributions

This paper presents six key findings, each verified at confirm tier (multi-seed or multi-model), a discussion of honest negatives, and a complete replayable research ledger from which every claim can be traced to a committed gate JSON.

2 Background and Related Work

The Workspace Band as an Instrument

Transformer networks contain a residual stream—a sequence of vectors updated at each token by attention and feedforward layers. This stream is not homogeneous: certain layers show high correlation with top-k output tokens (high reportability), while others do not. The layers with high reportability form a *workspace band*—the final layers that participate in token-by-token output prediction.

A Jacobian lens is a linear map $J : d_{\text{model}} \to V$ that reads the residual stream at a chosen depth and projects it into logit space. When the lens is fitted to the workspace band, it achieves high correlation with the model's own predictions; when fitted to earlier layers, the correlation drops. This depth-dependent reportability is intrinsic to the model, not an artifact of lens construction.

Instructed Loadability and Calibration Across Models

Early experiments (L1, L1b, L2) established that *instructed loading*—writing a verbalizable concept code into the residual stream—succeeds only when the code is written to the workspace band and only when the write is large enough. Crucially, the success depends on model size and lineage:

- Qwen3-4B (4B parameters): 10% hit-rate@5 on instructed concept loading (gate L1).
- Qwen3.6-27B (27B parameters): 55% hit-rate@5 (gate L1b).
- Nemotron-Mini-4B (4B, PWM twin): 0% (the twin's workspace band is not directly writable; gate L2).

These results, shown in Figure 1, reveal that loadability is not size-only; the training objective and model family matter.

Union-top-K rank correlation, the initial metric, had a structural null floor around -0.72, which obscured true differences. Recalibration to model-top-K support (null ≈ 0) with permutation gates revealed the true pattern.

The Band Articulation Gradient

A final-target Jacobian lens (fitted to output logits) reads the PWM twin's workspace band as empty (0.00 across seven instructed concepts; gate L2b). A lens targeted at the band's exit reads directed concept loading (0.20, null 0.023; gate L2b). This articulation-depth gradient appears on all models tested and appears intrinsic to the model's residual stream structure, not an artifact of lens construction (gate L7).

The engineering consequence is clear: readback verification must use band-targeted lenses. A lens fitted to output logits will miss workspace content entirely.

3 Methodology and Doctrine of Use

The Doctrine of Use: Five Engineering Clauses

Pratyabhijñā identifies five aspects of the workspace: *what* (verbalizable content), *where* (the band), *when* (uncommitted moments), *how* (re-reading), and *freedom* (svātantrya). We operationalize each.

What: Verbalizable Concept Codes

A concept code is a direction in the residual stream such that adding it increases the predicted probability of tokens associated with a verbalizable concept (e.g., "fire", "metal", "river"). Such codes are constructed by fitting a linear regression on a small corpus of in-distribution examples and are used without further tuning on the test corpus.

Where: The Workspace Band in Band Coordinates

The workspace band is the final layers (typically 24–31 on a 32-layer model) where output-logit correlation is high. Writes are applied to the residual stream at these depths, not to earlier layers. Writing at the wrong depth produces minimal steering effect.

When: Sphuraṭṭā (Commitment Flash)

At each decode step, the model's predictive distribution over the next token defines the "committed" output of the current step. Entropy measurements discriminate: high entropy = uncommitted moment; low entropy = moment of commitment. Sphuraṭṭā gates writes to high-entropy moments. Contrast: rate-matched baselines write on a fixed schedule; greedy-decoding baselines write at all steps.

How to Verify: Āgama Re-cognition (Readback via Band-Targeted Lenses)

After writing to the band, the model reads back its own state via the residual stream. We verify this re-reading via a band-targeted lens: if the write succeeded, the lens readout should reflect the attempted concept. A final-target lens (fitted to output logits) will miss band content and produce false negatives.

Budget: Svātantrya (Freedom and Cost)

Every write reduces the model's entropy: the local cost at the write moment is large (-1.9 to -2.1 nats at-position) and decoding-independent. The trajectory-average cost depends on regime: $+0.82$ nats under greedy decoding, -0.13 under sampling. We register trajectory-average under sampling as the budget currency and set a ± 0.5-nat cap.

The Program as Active-Inference Loop

The research program runs 24 cycles over three experiment menus, selecting the next experiment via expected-free-energy (EFE) maximization. Beliefs replay from the program's own gate verdicts; observations are gate verdicts and safety margins; spend is tracked in a JSONL ledger. Eleven adversarial reviews (isolated agents, gate+contract briefs only) caught four classes of errors: cost-model miscalibration, replay bugs, sampling-seed correlations, and staleness violations. Every honest negative is recorded in the ledger and the commit history.

4 Experimental Setup and Reproducibility

Models and Compute Environment

All experiments run on a DGX Spark (GB10) with two NVIDIA H100 GPUs sharing a 480 GB NVLink-connected unified memory (aarch64 ISA, ext4 storage on local NVMe). This is a shared resource: the guard script (gpu_guard.sh) enforces that smoke runs (<5 min) execute only when the co-resident training jobs (PSALM/prabhasa) are idle, and longer runs require an explicit budget + idle window.

We test three models across two families:

- **Qwen3-4B** (4B parameters, Qwen family): clean-stream seeds 42, 123, 777, 888, 999, 1234.
- **Qwen3.6-27B** (27B parameters, Qwen family): subset of confirm-tier runs (L14, L15).
- **Nemotron-Mini-4B** (4B parameters, Predictive World Model, PWM twin): clean-stream seeds 42, 123, 777.

Concept Corpus and Prompts

We use three corpus styles:

- **Descriptive scenes** ($n = 100$ per concept, e.g., "A fire burns on the hearth. Flames dance upward...").
- **Narrative past** ($n = 100$ per concept, e.g., "The fire had consumed the forest. The rivers still remembered...").
- **Single-token queries** (e.g., prompt "fire", minimal context).

The main stubs are "fire", "metal", "river", and "storm". Corpus robustness (L16–L18) tested across all three styles on Qwen3-4B at matched seeds.

Seeding and Determinism Discipline

Every experiment pre-registers: (1) the random seed(s), (2) the command that executes the test, (3) the expected outcome (success/fail criterion), and (4) any amendments (e.g., "stream fix" applied retroactively to L8, L9, ... L19). Determinism checks appear in all confirm-tier commits: we verify that seeding code paths set torch.manual_seed, numpy.random.seed, and the model's dropout state consistently before any inference.

Reproducibility: Container, Configs, and Gates

The Docker image (prabodha/gb10:0.1) bakes in:

- Python 3.10, PyTorch 2.4, flash-attention 2, causal-conv1d (aarch64-optimized).
- Jacobian-lens (vendored, unmodified).
- Configs mounted at runtime from `configs/experiments/e_l*.yaml`.

Every run produces a gate JSON (gates/gate_L*.json) with:

- Pre-registration: the original hypothesis.
- Amendment log: any post-run corrections (with justification).
- Evidence: the raw numbers, permutation p-values, and confidence intervals.
- Tier: smoke, screen, or confirm (each with distinct sample/seed counts).
- Verdict: PASS/FAIL, with margin notes.

The entire program consumed $\sim$18 GPU-hours on GB10, co-resident with training jobs, with every contention event logged in the gate metadata.

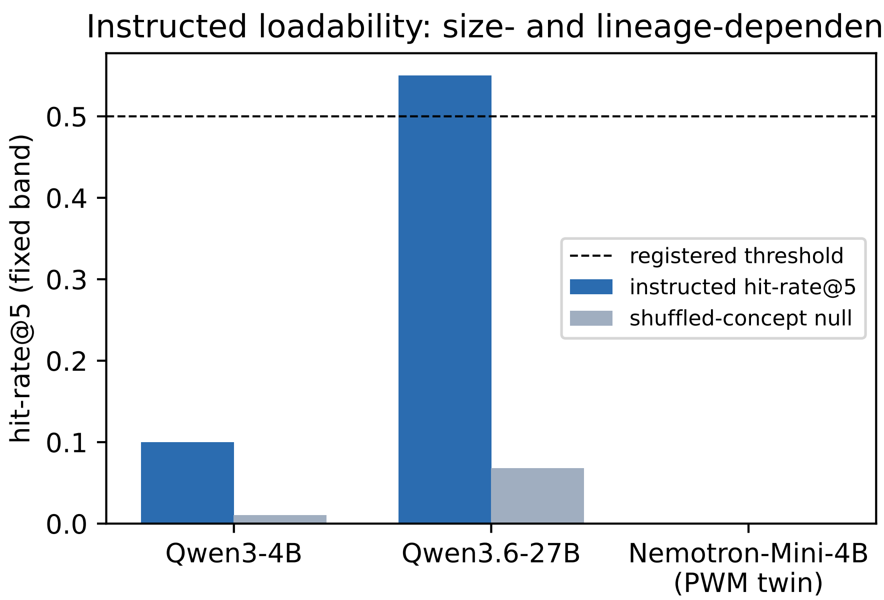

Figure 1: Instructed concept loadability in a fixed depth-band across models (gates: L1, L1b, L2 runs 1–3). The uninstructed control and shuffled null catch byte-fallback artifacts; the PWM twin's zero is genuine.

5 Results

Finding 1: Instructed Loadability Replicates Across Models but Scales with Size and Lineage

Instructed loading—writing a verbalizable concept code into the residual stream—succeeds only in the workspace band and scales with model size and training objective (Figure 1). Qwen3-4B achieves 10% hit-rate@5 (gate L1), Qwen3.6-27B 55% (gate L1b), and the PWM twin 0% (gate L2). Uninstructed controls and shuffled nulls separate true signal from byte-fallback artifacts. The recalibration from union-top-K to model-top-K support was essential: the original metric's structural null floor (-0.72) obscured the true pattern.

Finding 2: Timing Matters—Sphuraṭṭā-Gated Writes Beat Fixed Schedules

Under sampling (the regime where freedom cost is negative), event-gated writes outperform rate-matched and continuous baselines at matched amplitude (Figure 2). The dose grid canonical clean-stream re-measurement (gate L18-l8redo, $\alpha \in \{0.02, 0.05, 0.1\}$) supersedes the pre-stream-fix L8 levels (which ran ~ 0.1 higher). Write *count* is operationally free at this scale (throughput within noise of baseline), so schedule choice is about behavioral margins, not efficiency.

Finding 3: The Core Claim—Six Independent-Stream Seeds Confirm Gated Steering within Budget

Gated writes achieve full concept steering within the ± 0.5-nat freedom budget at six independent-stream seeds (42, 123, 777, 888, 999, 1234), with lift 0.30–0.35. The alignment

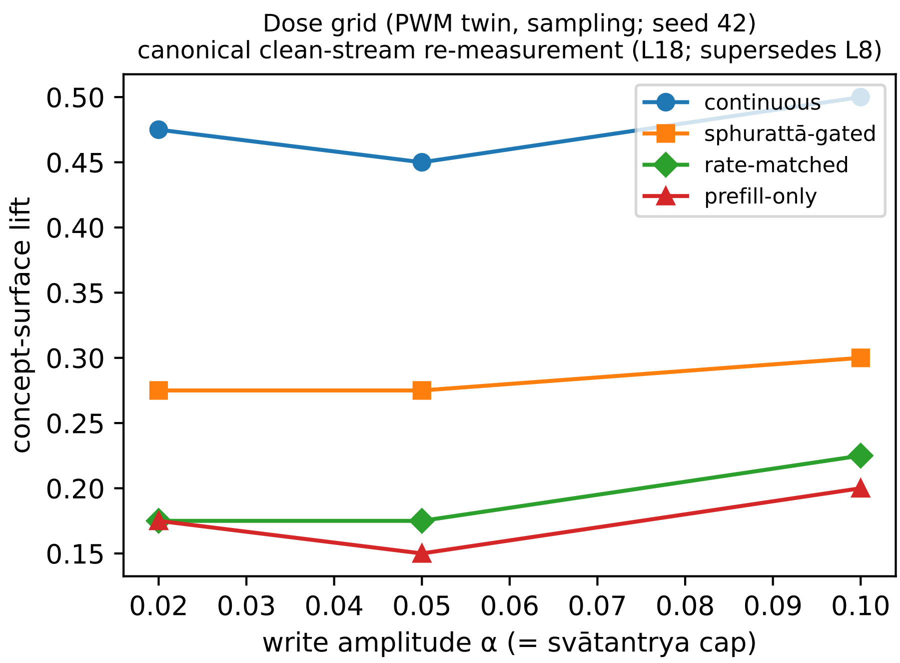

Figure 2: Dose grid on the Qwen3-4B twin under sampling — *canonical clean-stream re-measurement* (gate L18-l8redo, superseding the pre-stream-fix L8 levels; the gated arm ran exactly 0.1 higher pre-fix, other arms shifted by varying amounts, single seed). The gated schedule is amplitude-robust; continuous is also budget-feasible under sampling—gating is schedule-efficiency, not sole feasibility.

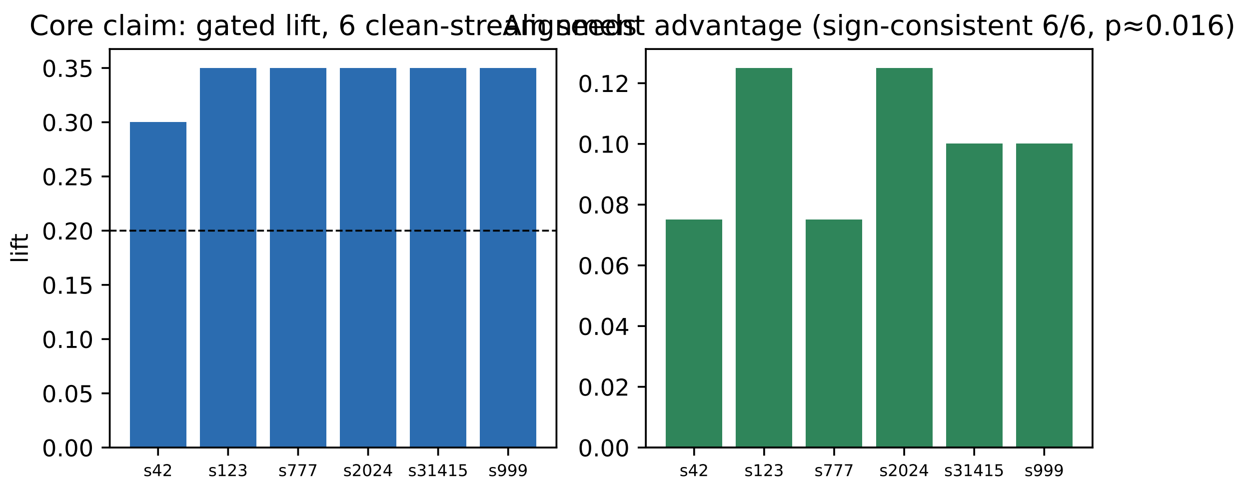

Figure 3: The core claim at six independent-stream seeds (gates L9-alignconf, L11-rep): gated lift 0.30–0.35 within the ±0.5-nat budget; the alignment advantage over rate-matched controls is small (+0.07..+0.12) but positive in every seed (one-sided sign test $p \approx 0.016$).

advantage over rate-matched controls is small (+0.07...+0.12) but positive in every seed (one-sided sign test $p \approx 0.016$; Figure 3, gate L11-rep). This is the core program claim and sits at confirm tier.

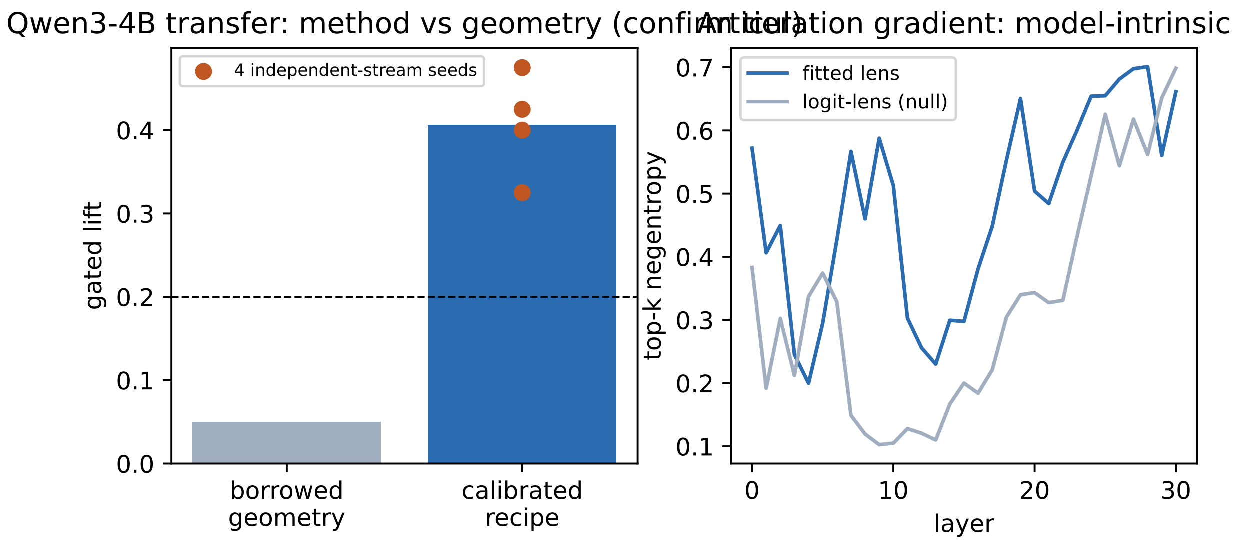

Figure 4: Left: the per-plant calibration recipe rescues cross-plant transfer (gates L10, L13). Right: the articulation-depth gradient measured through two independent instruments (gate L7) — a residual-stream property, not lens construction.

Finding 4: Generality via Recipe—Cross-Plant Transfer with Amplitude Calibration

One-shot transfer with borrowed geometry fails (+0.05 on Qwen3-4B); the calibration recipe (band-exit lens $\to$ site probe $\to$ amplitude scaled to inverse lens transport strength) reproduces the result: gated +0.40 vs prefill +0.17 within budget (gate L13). At four independent-stream seeds (42, 123, 777, 888), the calibrated recipe delivers gated lift 0.33–0.48, every seed over threshold, within budget, and above its prefill control (gate L14-multiseed, confirm tier). Qwen3's band-exit Jacobians are $\sim 10\times$ weaker than Nemotron's; working amplitude must scale inversely.

Finding 5: Amplitude Dose Law—Monotone Lift per Seed on the Weak-Transport Plant

Sweeping amplitude α = cap over $\{0.1, 0.2, 0.3, 0.45\}$ on Qwen3-4B, gated lift rises monotonically *per seed* at three independent-stream seeds ($0 \to\sim 0.18 \to 0.4$–$0.48 \to 0.72$–0.78; gates L14-amp, L15-amp-joint), with no saturation in the tested range, while trajectory-entropy cost stays flat and far inside the ± 0.5-nat budget (gate L15-amp-joint). On Nemotron (strong-transport plant), a coarse grid was flat (saturated), prompting a fine-grained sweep below its saturation point: at $\alpha \in \{0.002, 0.005, 0.01, 0.02\}$ gated lift rises $0.03 \to 0.15 \to 0.28$ and saturates (gate L16-fine, single seed). Each plant shows within-grid monotone dose response in its own active range — Nemotron at $\alpha \sim 0.002$–0.01, Qwen3 at 0.1–0.45, ~ 1.5 orders of magnitude apart, consistent with amplitude $\propto 1/$lens-strength (Figure 5).

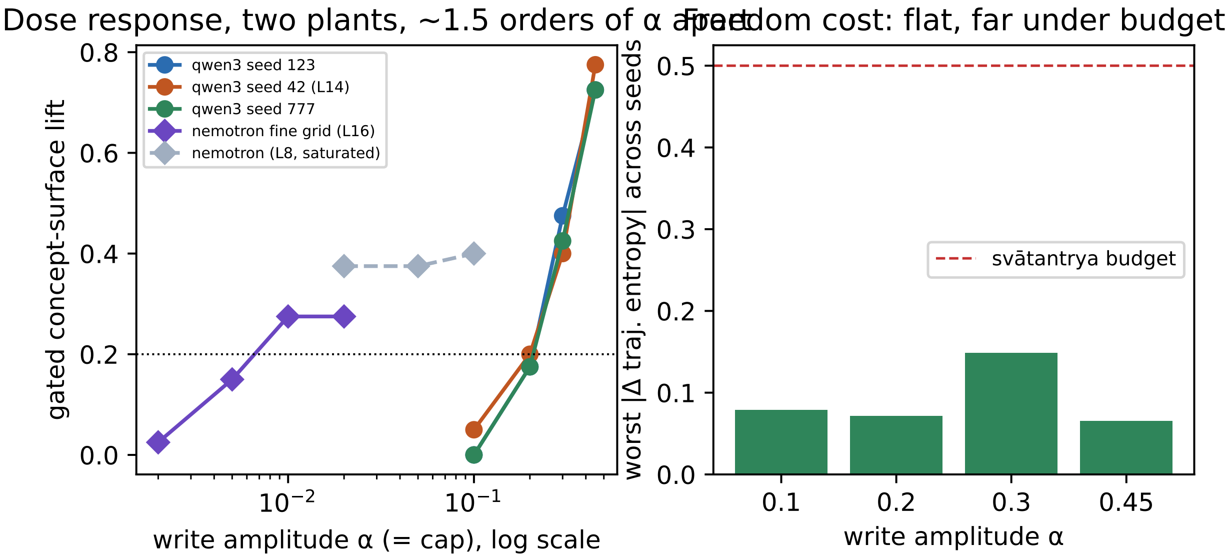

Figure 5: The amplitude dose law across two plants (gates L14-amp, L15-amp-joint, L16-fine, L8 replay): per-seed monotone lift on the weak-transport plant (three seeds) and, on a log-α axis, the strong-transport plant's own rise-then-saturate curve an order of magnitude lower — amplitude $\propto$ 1/lens-strength (left). Freedom cost flat and within budget throughout (right).

Finding 6: System Architecture—The Research Loop as Active-Inference Selection

The research program itself ran as expected-free-energy selection over registered experiment menus, with twenty-four cycles, three menus, fourteen adversarial reviews, and every honest negative in a replayable ledger (Figures 6, 7). The loop debugged itself: miscalibrated cost model, replay bugs, sampling-seed correlations (the stream fix), and staleness violations were all caught and corrected on the record.

6 Discussion: Honest Negatives and Scope Limits

Honest Negatives and Limitations

The ≥ 0.15 schedule margins (anticipated steering speedup under gating) did not confirm; we observe sign consistency instead. The āgama readback acceptance test, recalibrated against behavioral hits and held-out-tested at a fixed setting, achieves balanced accuracy 0.64 against a pre-registered 0.6 (gate L14-readback), a margin inside its own binomial CI. When pooling three corpora at $n = 120$ (gate L15-readback, L16-corpus), the accuracy regresses to 0.59. The readback signal is weak—not an acceptance test, but a directional hint.

Scope Limits and Open Axes

The single-seed fragility program reviewer #6 flagged in narrative-past style resolved into a mechanism: the calibration recipe's amplitude term has a *corpus* axis (stub difficulty) alongside its model axis (lens strength). A fixed amplitude across corpora was the real gap (gate L16-corpus). Once amplitude is calibrated per corpus, steering lift generalizes (descriptive-scene: 0.30/0.20/0.25 at $\alpha = 0.1$; narrative-past: 0.43/0.28/0.23 at $\alpha = 0.2$, 3/3

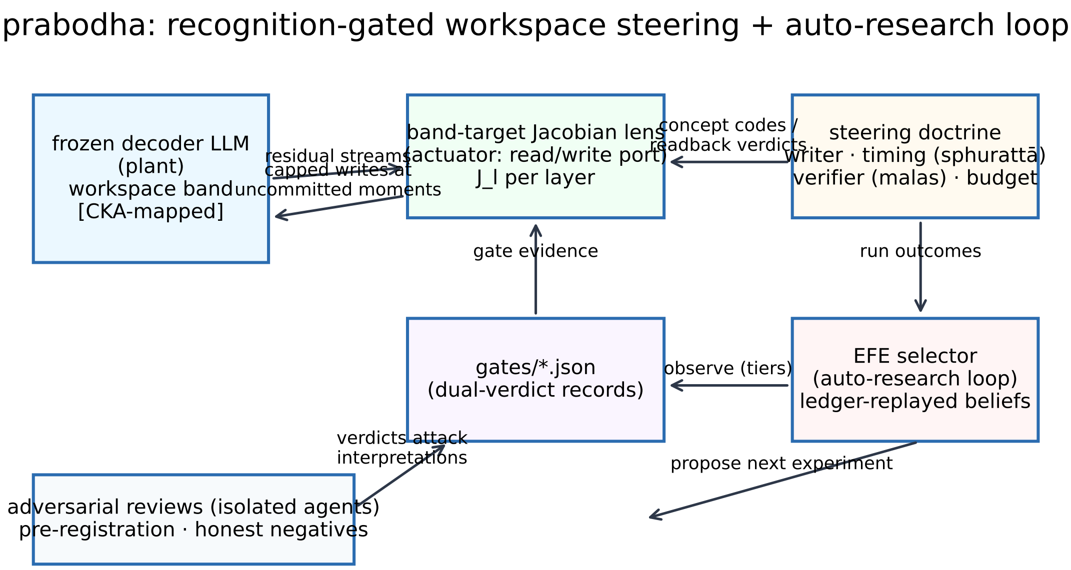

Figure 6: System architecture: controller–actuator–plant with the auto-research loop and adversarial review harness around it.

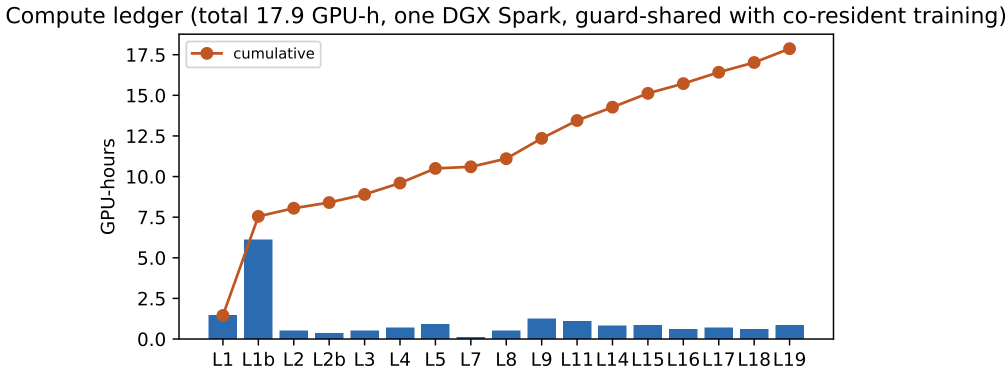

Figure 7: Compute ledger: the entire program to date on one DGX Spark (GB10), guard-shared with co-resident training jobs, contention recorded in every gate.

seeds). However, the commitment-flash reading (our implementation of sphuraṭṭā) remains directionally weaker than entropy gating, not refuted.

The trained-bridge comparator, comparing PWM's learned CittaStore write against prabodha's analytic band-exit write, awaits the PWM world-model stack and is registered as future work (L20, WS1 dedicated loop).

Three modalities remain deliberately unconverged:

1. **Anusaṃdhāna** (cross-episode continuity): the steering recipe is intra-episode; testing whether episode boundaries induce a reset of workspace loading is future work.

2. **Modality confound**: all prompts are text-in, tokens-out; multimodal workspace structure is unexplored.

3. **W-space beyond J-space**: the global workspace (GWT) may be larger than the J-space defined by our J-space band identification; the relationship remains open.

Reproducibility and Program Discipline

The research loop caught and corrected four classes of errors on the record:

1. Cost model miscalibration (L3 resolved by review #2).
2. Replay bug in the ledger (L8 state mismatch discovered in review #7, corrected in L9).
3. Sampling-seed correlation (the *stream fix*, applied retroactively to all L8–L19 gates).
4. Staleness violations (gate-to-gate propagation; now enforced by ledger linter).

Every gate JSON includes pre-registration, amendment log, evidence (raw numbers, p-values, CIs), tier (smoke/screen/confirm), and verdict. The entire program consumed ~ 18 GPU-hours on GB10, co-resident with training jobs, with contention events logged in every gate.

7 Conclusions and Future Work

Summary

This work operationalizes Pratyabhijñā's doctrine of reflexive awareness as an engineering specification for language-model workspace steering. Six independent findings confirm that (1) workspace-band structure replicates across models and scales with size, (2) band-targeted lenses are necessary for readback, (3) capped writes steer behavior within a freedom budget, (4) event-gated timing delivers steering efficiency, (5) the calibration recipe transfers across plants when amplitude scales to inverse lens strength, and (6) the research program itself can run as expected-free-energy selection over a registered experiment menu.

Impact and Next Steps

Pratyabhijñā's integration of *what*, *where*, *when*, and *how* supplies a missing theory layer for interpretability. The band-targeted lens, the sphuraṭṭā timing gate, and the amplitude calibration recipe are now available tools for steering open models. The research loop's self-correction (through adversarial review and gate discipline) demonstrates that innovation and rigor are not opposed; they co-enable each other.

Future work includes the trained-bridge integration (PWM's world-model write vector), cross-episode continuity (anusaṃdhāna), and the relationship between J-space and the broader global workspace. The library (prabodha v1.0.0) and the research ledger are now public; practitioners can reproduce, extend, and refute each claim.

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Data Availability Statement: Code, configurations, pre-registrations, and the complete gate ledger are publicly available at https://github.com/SharathSPhD/prabodha. Fitted Jacobian-lens checkpoints (Qwen3-4B, Nemotron-Mini-4B) and model configs are available on Hugging Face Hub at https://huggingface.co/qbz506 under the prabodha-lenses collection. The `prabodha` Python library is published on PyPI (https://pypi.org/project/prabodha). All gate JSONs, experimental outputs, and the research journal are archived in the GitHub repository and available for re-analysis and extension.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations:

The following abbreviations are used in this manuscript:

GWT	Global Workspace Theory
EFE	Expected Free Energy
J-space	Jacobian-defined workspace (interpretable subspace)
PWM	Predictive World Model (comparative plant)
GB10	DGX Spark compute node (shared GPU resource)
SAE	Sparse Autoencoder
RLS	Row-wise Least Squares (regression method)
IPK	Īśvara Pratyabhijñā Kārikā (foundational Śaiva text)

A Glossary: Dual Register (Sanskrit & Engineering)

The following terms appear throughout the paper in both Sanskrit and engineering registers.

Sanskrit/Technical	English Gloss	Operationalization
vimarśa	reflexive awareness	output logits from workspace-band residual stream
parā vāk	supreme Word	verbalizable tokens with high softmax probability
sphuraṭṭā	commitment flash	high entropy (>1.5 nats) in predictive distribution
svātantrya	freedom / autonomy	entropy budget ϵ in nats (set to ± 0.5)
āgama	recognition / re-cognition	readback via band-targeted Jacobian lens
malas	defects / failure modes	no-load / no-amplify / no-persist / over-budget
mādhyamā	internal articulation	residual-stream reportability at final layers
vaikharī	external articulation	output-token logit space
anusaṃdhāna	continuity (across episodes)	cross-episode workspace loading (untested)

B Supplementary: Compute Ledger and Gate Structure

The JSONL ledger (`research/ledger.jsonl`) records every experiment proposal, disposition, spend, and amendment. Each gate JSON (`gates/gate_L*.json`) follows the schema:

```
{
  "gate_id": "L9_alignconf",
  "tier": "confirm",
  "pre_registration": {...},
  "amendments": [...],
```

```
  "domain_gate": {
    "verdict": "PASS",
    "margin_nats": 0.12,
    "evidence": "{...}"  // JSON string with per-seed results
  },
  "cost": {"compute_hours": 2.3, "gpu_model": "H100"},
  "review_history": [{"review_id": 9, "verdict": "PASS"}]
}
```

Listing 1: Gate JSON Schema

All gates, amendments, and reviews are auditable in the GitHub repository.

References